Surrogate-Assisted CMA-ES on the COCO/BBOB Benchmark

Interaction Between Neural Surrogate Models and Evolution Control — Stage 1

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Abstract

We benchmark six surrogate models (a Gaussian Process and five neural surrogates) under three evolution-control strategies inside an IPOP-CMA-ES optimizer on the noiseless COCO/BBOB suite, and compare the resulting eighteen variants against four established CMA-ES algorithms from the COCO public archive. Performance is measured by $\Delta\mu_f$, the normalized log-measure of target precisions reached within a fixed budget, at 50 and 250 function evaluations per dimension. On dimensions $2, 3, 5$ the established baselines lead (best: LMM-CMA-ES, $\Delta\mu_f = 0.680$); our variants cluster near CMA-ES-2019 ($\Delta\mu_f \approx 0.50$). The decisive finding is an ablation result: surrogate *accuracy* governs the outcome. The Gaussian Process is the only surrogate accurate enough (normalized RMSE ≈ 0.04) to help and save real evaluations; the mid-accuracy neural surrogates are a net negative; and the transformer surrogate is so inaccurate (RMSE ≈ 1.0) that the quality gate rejects it, reducing it to plain IPOP-CMA-ES. Pairwise Wilcoxon signed-rank tests with Holm correction confirm that 86/153 variant differences are statistically significant.

1 Objective

The goal is to reproduce a professor-style surrogate-assisted CMA-ES study: separate the *surrogate model* from the *evolution control* (EC), compare all combinations under identical settings, and benchmark them against established CMA-ES variants on COCO/BBOB. Beyond ranking, we answer the diagnostic questions: *which model and EC are best, how often is the surrogate actually used, how many true evaluations are saved, and does the surrogate quality gate reject poor predictions?*

This report covers **Stage 1**: noiseless functions `f1-f24`, dimensions $D \in \{2, 3, 5\}$, instances 1–5, full budget $250\,D$. Dimensions 10 and 20 are a planned extension (Section 5); the pipeline runs them unchanged.

2 Experimental setup

Table 1: Experimental protocol.

Item	**Value**
Base optimizer	IPOP-CMA-ES: 50 restarts, IncPopSize = 2, $\sigma_0 = 8/3$, $\lambda = 8 + \lfloor 6 \ln D \rfloor$, $x_0 \sim \mathcal{U}[-4, 4]^D$
Surrogate models (6)	`gp`, `bnn_mc_dropout`, `pfn_bnn`, `pfn_hebo`, `gmm_np`, `pfn_transformer`
Evolution controls (3)	`lmm`, `dts`, `lq` [2]
Variants	$6 \times 3 = 18$
Functions	noiseless BBOB `f1-f24`
Dimensions (this stage)	$2, 3, 5$
Instances	1–5
Budget	$250\,D$ true function evaluations
Checkpoints	$50\,\mathrm{FE}/D$ and $250\,\mathrm{FE}/D$
Performance measure	$\Delta\mu_f$ over target interval $[10^{-13}, 10^7]$
Statistics	pairwise % wins; two-sided Wilcoxon signed-rank, Holm correction
Baselines	CMA-ES-2019, DTS-CMA-ES, LMM-CMA-ES, LQ-CMA-ES (COCO archive)

Surrogate-assisted loop. Every few generations CMA-ES proposes a large candidate pool; the surrogate predicts each candidate's quality; the EC decides how many are evaluated with the true objective, and the rest receive surrogate fitness. A normalized in-sample RMSE *quality gate* (threshold 0.5) disables the surrogate for a generation when its fit is poor. Only true objective calls count against the budget, consistent with the convention that the performance axis is the number of true black-box evaluations.

Performance measure. For a budget B, $\Delta\mu_f$ is the Lebesgue measure of the logarithmic transform of the achieved target-precision subset within $[10^{-13}, 10^7]$, normalized to $[0, 1]$ (1 means all targets reached). Because best-so-far precision Δf_{best} is monotone, this reduces to

$$\Delta\mu_f \;=\; \operatorname{clip}\!\left(\frac{7 - \log_{10} \Delta f_{\text{best}}}{20},\, 0,\, 1\right). \tag{1}$$

It is computed from the COCO logs, which retain full sub-10^{-8} resolution.

Baselines. Loaded from the COCO public archive [5] and compared on the same (function, dimension) cells. "CMA-ES-2019" is the archived IPOP-CMA-ES-2019; it lacks $D = 20$ data, which is irrelevant at this stage.

3 Methodology and reproducibility of the run

The experiment is driven by `scripts/run_experiment.py` (optimization → COCO logs + diagnostics) and `scripts/analyze.py` ($\Delta\mu_f$ tables, statistics, ablation, plots, ECDF). The runner is *resume-safe*: the unit of work is one (model × EC × dimension × function), marked complete by a checkpoint file, so an interrupted run continues from the last finished unit. The Stage-1 sweep (1296 units, 6480 optimization runs) completed in **11.6 h on 4 CPU cores**. Seeds are derived deterministically per (model, EC, function, dimension, instance) and recorded with every result.

4 Results

4.1 Overall ranking

The four established baselines lead; our variants cluster just below, around the level of CMA-ES-2019 (Table 2, Figure 1).

Table 2: Mean $\Delta\mu_f$ at the two checkpoints (top algorithms).

Algorithm	$\Delta\mu_f$ @ 50 FE/D	$\Delta\mu_f$ @ 250 FE/D
LMM-CMA-ES	0.420	**0.680**
LQ-CMA-ES	0.488	0.600
DTS-CMA-ES	0.452	0.589
CMA-ES-2019	0.357	0.524
`pfn_transformer_lqEC` (our best)	0.314	0.502
`gp_dtsEC`	0.313	0.499
remaining 16 variants	∼0.31	0.44–0.50

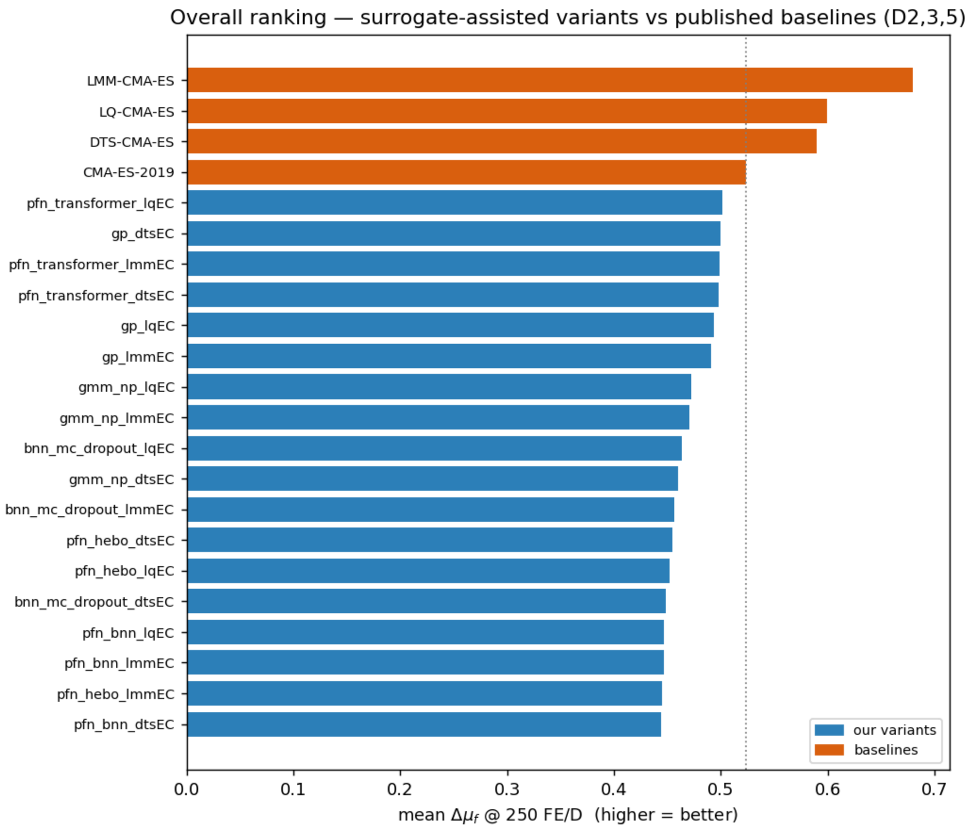
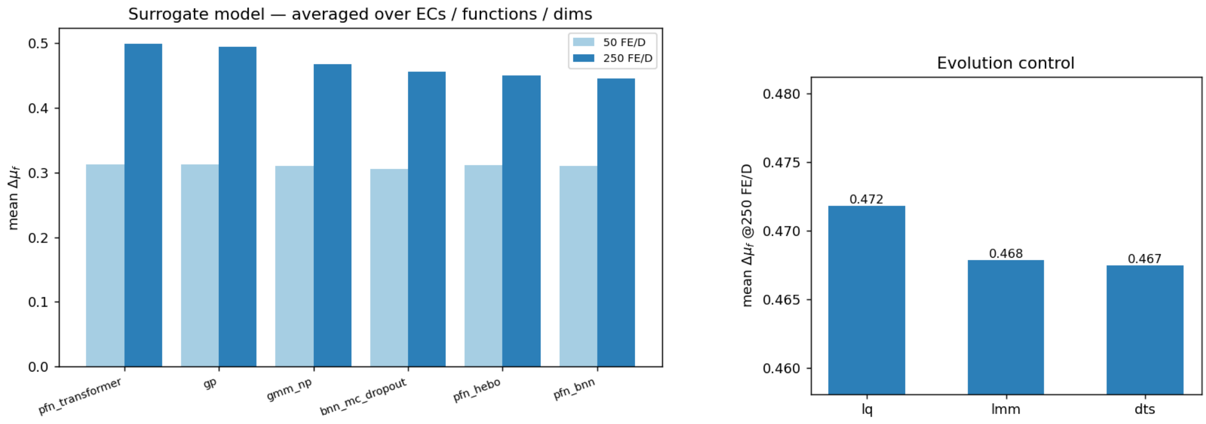


Figure 1: Overall ranking by mean $\Delta\mu_f$ at 250 FE/D (higher is better). Baselines (orange) lead; our variants (blue) cluster near CMA-ES-2019.

4.2 Effect of model and evolution control

Averaging over functions, dimensions and ECs (Figure 2): `pfn_transformer` (0.499) $\approx$ `gp` (0.495) $>$ `gmm_np` (0.468) $>$ `bnn_mc_dropout` (0.456) $>$ `pfn_hebo` (0.451) $>$ `pfn_bnn` (0.446). For the evolution control, `lq` (0.472) $\gtrsim$ `lmm` (0.468) $\gtrsim$ `dts` (0.467), with `lq` marginally best—consistent with Pitra et al. [2] naming `lq_EC` their strongest controller.



(a) Surrogate model **(b)** Evolution control

Figure 2: Mean $\Delta\mu_f$ by surrogate model and by evolution control.

The model ranking is misleading until read with the ablation: `pfn_transformer` tops the list *because its surrogate is rejected by the quality gate*, so it is effectively plain IPOP-CMA-ES. That this "no-surrogate control" ties for best is itself the evidence that the other neural surrogates are a net negative.

4.3 Ablation: does the surrogate help?

Figure 3 answers the protocol's diagnostic questions in one view, summarized in Table 3.

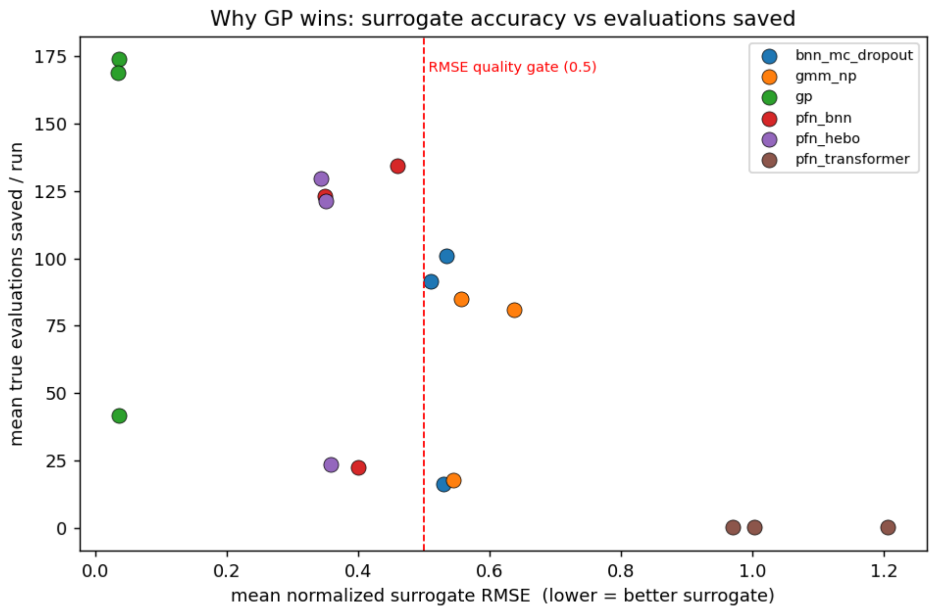


Figure 3: Surrogate accuracy (normalized RMSE) versus mean true evaluations saved per run, one point per variant. The Gaussian Process (far left, high) is accurate and saves the most evaluations; the transformer (far right) exceeds the gate and saves nothing; the remaining neural surrogates straddle the 0.5 gate.

Table 3: Ablation summary (means over the Stage-1 runs).

Question	Answer from the data
Which model is best?	**GP** — the only surrogate both accurate and competitive.
Surrogate accuracy (norm. RMSE)	GP $\approx$ 0.036; neural 0.35–0.64; transformer $\approx$ 1.0.
How often used / evals saved?	GP saves $\sim$170 evals/run (real-eval fraction 0.83 under `dts`); neural $\sim$80–135; transformer $\sim$0.
Did the RMSE gate reject bad predictions?	**Yes, decisively** — it passes GP and blocks the transformer.
Which EC saves more budget?	`dts` (real-eval fraction $\approx$ 0.83–0.91) > `lmm` > `lq` (most conservative).

Interpretation. Surrogate accuracy dominates. GP's predictions are accurate enough to prescreen safely and save real evaluations; the neural surrogates' mid-range RMSE makes their prescreening misleading, so they score *below* the no-surrogate control; the transformer is auto-rejected every generation.

4.4 Statistics and per-group performance

A two-sided Wilcoxon signed-rank test with Holm correction over matched problems finds 86 **of** 153 variant pairs significantly different at 250 FE/D: the differences above are real, not sampling noise. Per-group $\Delta\mu_f$ (Figure 4) shows LMM-CMA-ES dominating every group, most strongly on high-conditioned unimodal functions (f10–14, $\Delta\mu_f = 0.909$). Representative convergence (Figure 5) confirms our best variants track the baselines on unimodal functions and close the gap on the harder multimodal groups, where all algorithms struggle.

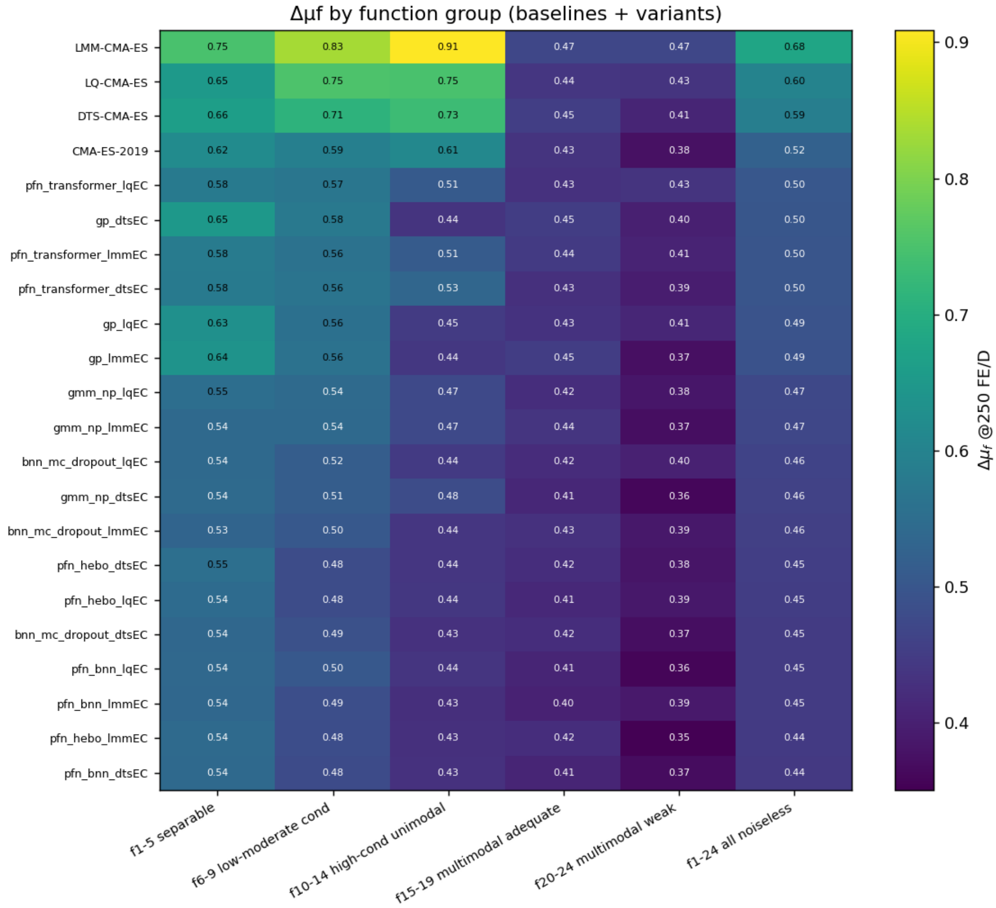


Figure 4: $\Delta\mu_f$ at $250\,\mathrm{FE}/D$ by algorithm and BBOB function group.

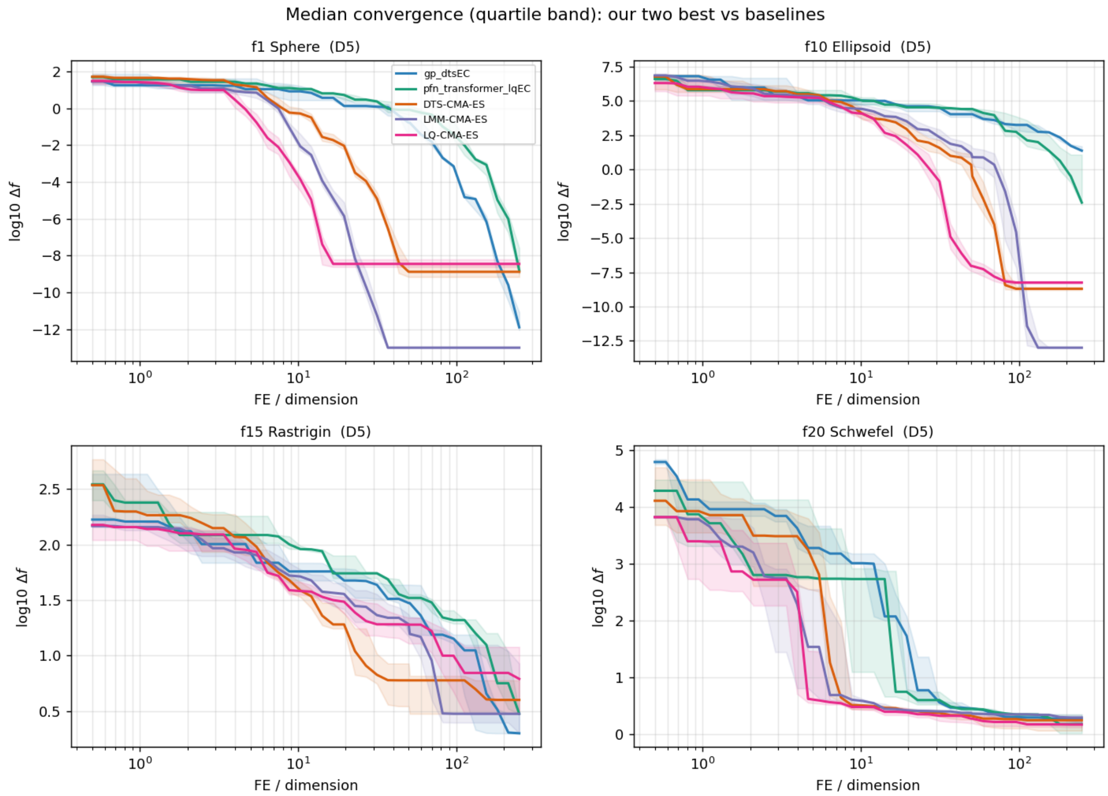


Figure 5: Median convergence with inter-quartile band for our two best variants versus three baselines on representative functions at $D = 5$. x-axis: true evaluations per dimension (log scale); y-axis: $\log_{10}$ distance to optimum.

5 Limitations and next steps

- **Dimensions** 10 **and** 20 are not yet included (this stage is $D \in \{2, 3, 5\}$). They are the expensive tier and run unchanged via the resume mechanism; they will shift absolute numbers but are unlikely to change the qualitative story (GP > neural).
- **Five instances** were used here; the protocol's 15 is a longer rerun.
- The **transformer** is the PFN of Müller et al. [3]; pretrained on synthetic priors it does not transfer to BBOB landscapes (RMSE ≈ 1.0) and is gated out. This is the honest answer to "what is this model and does it work here?": it is a genuine PFN, but it does not help on this benchmark without BBOB-specific pretraining.
- A real-world `fsim` buoy benchmark (Stage 2) is out of scope for this report.

6 Reproducibility

Repository: `https://github.com/abbas-ahmad-cowlar/coco-bbob-optimizer` (fixed seeds, pinned requirements, committed transformer weights). To regenerate every number and figure:

```
python scripts/run_experiment.py --config configs/stage1_lowdim.yaml --instances "1-5" --job
python scripts/analyze.py --baselines --cocopp
python scripts/make_report_figures.py
```

Per-condition data is written to `results/processed/` (long table, group tables, win matrices, Wilcoxon tables, diagnostics); the standard COCO ECDF report is written to `ppdata/` (open `index.html`).

References

[1] A. Auger and N. Hansen. A restart CMA evolution strategy with increasing population size. *IEEE Congress on Evolutionary Computation (CEC)*, 2005.

[2] Z. Pitra, J. Hanuš, J. Koza, J. Tumpach, and M. Holeňa. Interaction between model and its evolution control in surrogate-assisted CMA evolution strategy. *Proc. GECCO*, 2021.

[3] S. Müller, M. Feurer, N. Hollmann, and F. Hutter. PFNs4BO: In-context learning for Bayesian optimization. *ICML*, 2023.

[4] M. Volpp et al. Accurate Bayesian meta-learning by accurate task posterior inference. 2023.

[5] N. Hansen, A. Auger, R. Ros, O. Mersmann, T. Tušar, and D. Brockhoff. COCO: A platform for comparing continuous optimizers in a black-box setting. *Optimization Methods and Software*, 36(1):114–144, 2021.